

# Bluestock Fintech

## Analyst Guide — Nifty100 Financial Analysis

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### Purpose

This guide explains how to use the Bluestock Fintech Nifty100 Financial Analysis platform. It covers the Streamlit dashboard, the Stock Screener, peer comparison, sector analysis, PDF tearsheet generation, the FastAPI REST API, and common troubleshooting steps.

The platform provides financial analytics for 92 Nifty 100 companies, backed by a SQLite database containing profit-and-loss statements, balance sheets, cash flow data, financial ratios, market capitalisation, and peer-group mappings.

### Contents

1. Getting Started — Environment, installation, and startup
2. Dashboard Overview — Navigation and the eight screens
3. Home Dashboard — KPIs, sector view, top companies
4. Company Profile — Selection, tabs, financial information
5. Stock Screener — Presets, filters, sorting, pagination, CSV export
6. Peer Comparison — Relative metrics and interpretation
7. Financial Trends — Historical performance charts
8. Sector Analysis — Sector-level aggregates
9. Capital Allocation — Investment strategy insights
10. Reports — PDF tearsheets and report generation
11. API Guide — curl examples for REST endpoints
12. Troubleshooting — Common issues and solutions
13. Operational Reference — ETL, testing, and command reference

# Getting Started

## Environment

The platform runs on Python 3.12 with the following key dependencies:

- Streamlit 1.58.0 (web dashboard)
- FastAPI 0.116.1 (REST API)
- pandas 2.3.3 (data manipulation)
- numpy 1.26.4 (numerical computing)
- scikit-learn 1.3.0 (clustering)
- Plotly 5.18.0 (charting)
- ReportLab 5.0.0 (PDF generation)
- SQLite3 (database, built into Python)

## Installation

Create and activate a virtual environment:

```
python -m venv .venv
source .venv/bin/activate (Linux/macOS)
.venv\Scripts\activate (Windows)
```

Install dependencies:

```
pip install -r requirements.txt
```

## Startup

To start the Streamlit dashboard:

```
streamlit run src/dashboard/app.py
```

To start the FastAPI backend:

```
uvicorn src.api.main:app --reload
```

The dashboard opens at <http://localhost:8501> and the API at <http://localhost:8000>.

API documentation is available at <http://localhost:8000/docs>.

# Dashboard Overview

The Streamlit dashboard provides eight navigation screens accessible via the sidebar menu. Each screen serves a distinct analytical purpose.

| Screen             | Description                                                |
|--------------------|------------------------------------------------------------|
| Home               | KPI overview with sector distribution and top companies    |
| Company Profile    | Detailed financial analysis for a selected company         |
| Screener           | Filter and rank companies by financial criteria            |
| Peer Comparison    | Compare a company against its peer group                   |
| Financial Trends   | Historical financial performance charts (5&ndash;10 years) |
| Sector Analysis    | Sector-level aggregation and comparison                    |
| Capital Allocation | Investment classification and strategy                     |
| Reports            | Report generation and tearsheet management                 |

Navigation: Use the sidebar on the left to switch between screens. Some screens require selecting a company or year before data loads.

# Home Dashboard

The Home screen provides a high-level overview of the Nifty 100 market. It displays six KPI cards, a sector distribution donut chart, and a top-5 companies table.

## KPI Cards

- Average ROE — mean return on equity across all companies
- Median P/E — price-to-earnings ratio median
- Median Debt/Equity — median leverage ratio
- Total Companies — count of companies in the dataset
- Median Revenue CAGR (5yr) — median 5-year revenue growth
- Debt-Free Companies — count of companies with zero or null debt

## Sector Distribution

A donut chart visualises the distribution of companies across broad sectors. Hover over segments for sector names and percentages.

## Top 5 Companies

A table ranks the top 5 companies by a composite quality score that weights profitability (35%), cash quality (30%), growth (20%), and leverage (15%).

## Year Selector

Use the year dropdown in the sidebar to view KPIs for a specific fiscal year.

# Company Profile

The Company Profile screen provides detailed financial analysis for a single selected company. Begin by selecting a company from the dropdown, then click 'Load Profile'.

## Company Selection

Use the 'Select Company' dropdown to choose from the 92 available companies. Company tickers are unique IDs (e.g., 'TCS', 'RELIANCE').

## Profile Header

The header displays the company name, ticker, sector, industry, and market capitalisation.

## Tabs

**Financial Ratios:** Annual financial ratios including ROE, ROCE, debt-to-equity, interest coverage, and more

**Cash Flow:** Annual cash flow data: operating, investing, and financing activities

**Capital Allocation:** Capital allocation metrics: ROE, debt-to-equity, free cash flow, and cash conversion ratio

**Raw Financials:** Placeholder for full P&L; and balance sheet data

## Key Ratio Cards

The Financial Ratios tab displays four key metric cards for the latest year: ROE, ROCE, Debt/Equity, and Interest Coverage.

# Stock Screener

The Stock Screener allows you to filter and rank the 92 Nifty 100 companies by up to 12 financial criteria. Results are scored by a composite quality metric and can be sorted, paginated, and exported.

## Preset Filters

- Quality Compounder
- Value Pick
- Growth Accelerator
- Dividend Champion
- Debt Free Blue Chip
- Turnaround Watch

## Filter Criteria

- ROE — Return on Equity %
- Debt to Equity — Leverage ratio
- Operating Profit Margin — OPM %
- Market Cap — Market capitalisation (crore INR)
- Free Cash Flow — FCF in crore INR
- Revenue CAGR — Compound sales growth
- PAT CAGR — Compound profit growth
- Dividend Yield — Annual dividend yield %
- PE Ratio — Price to earnings
- PB Ratio — Price to book
- Interest Coverage — Interest coverage ratio

## Sorting and Pagination

Results are sorted by the composite quality score by default. Use the sort dropdown to change the sort field and direction. The table supports pagination with configurable page size (default: 50 per page).

## CSV Export

Click the 'Download CSV' button below the results table to export the current filtered and sorted results as a CSV file.

# Peer Comparison

The Peer Comparison screen allows you to assess a company relative to its peer group. Percentile rankings show how the selected company ranks on key financial metrics within its peer cohort.

## Usage

1. Select a company from the dropdown.
2. Click 'Load Profile'.
3. The screen displays percentile rankings for ROE, NPM, Debt/Equity, FCF, PE, and PB against the company's peer group.
4. The overall peer score is the average of all available percentile rankings.

## Interpretation

A percentile above 50 indicates the company outperforms its peers on that metric. For Debt/Equity, PE, and PB, lower values are better, so percentile scores are inverted accordingly.

# Financial Trends

The Financial Trends screen displays multi-year historical charts for a selected company. Charts cover up to 10 years of revenue, net profit, ROE, and ROCE data.

## Charts

### Revenue & Net Profit

Grouped bar chart showing annual revenue and net profit over up to 10 years

### ROE & ROCE Trend

Dual-axis line chart showing annual ROE and ROCE trends

Use the company dropdown to select a company and view its historical performance. Missing years are excluded from the charts to maintain data integrity.

## Sector Analysis

The Sector Analysis screen aggregates financial metrics across all broad sectors. It displays company counts, average ROE, average ROCE, average net profit margin, average debt-to-equity, average PE ratio, and total sector market capitalisation.

## Sector-level Aggregates

Sector aggregates are computed across the most recent market capitalisation year. Use the sector table to compare sector performance side by side.



# Capital Allocation

The Capital Allocation screen provides insights into how companies allocate capital, including investment classifications and strategic recommendations.

## Capital Allocation Categories

### **Excellent**

Companies with superior capital allocation practices

### **Good**

Companies with strong capital allocation

### **Average**

Companies with acceptable capital allocation

### **Weak**

Companies with suboptimal capital allocation

### **Poor**

Companies with poor capital allocation

## Usage

Select a company and click 'Load Profile' to view its capital allocation classification. The badge colour indicates the quality of capital allocation.

## Reports

The Reports screen offers three tabs: Peer Comparison Report (Excel), Custom Report Builder, and Generated Reports listing.

### Peer Comparison Excel Report

Click 'Generate Peer Comparison Report' to create an Excel file with one worksheet per peer group. Each sheet includes company name, peer group, overall peer score, and all percentile metrics with conditional formatting (green for top 20%, yellow for middle 60%, red for bottom 20%).

### Custom Report Builder

Select multiple companies and metrics to build a custom report. Currently in preview mode.

### Generated Reports

Lists all previously generated reports in the Data/output directory.



# PDF Tearsheets

A tearsheet is a two-page A4 PDF summary for a single company, generated using ReportLab. It includes KPI tiles, financial trend charts, balance sheet composition, cash flow waterfall, pros/cons, and a capital allocation badge.

## What Is in a Tearsheet?

### Page 1

KPI tiles (ROE, ROCE, OPM, Debt/Equity, FCF Conversion, Revenue CAGR), Revenue & Net Profit bar chart, ROE & ROCE trend chart

### Page 2

Balance sheet composition stacked bar, cash flow waterfall, pros and cons list, capital allocation badge

## Generation Method

Tearsheets are generated programmatically using the reportlab library. The `generate_tearsheet` function in `src/reports/tearsheet.py` creates a 2-page PDF.

## Output Location

PDFs are saved to a specified output path. Example:

```
from src.reports.tearsheet import generate_tearsheet
generate_tearsheet("TCS", "output/tearsheets/tcs_tearsheet.pdf")
```

## Required Parameters

`company_id`: The company ticker (e.g., 'TCS'). must exist in the database.

`output_path`: File path where the PDF will be written.

# API Guide

The FastAPI backend exposes REST endpoints at <http://localhost:8000/api/v1/>. Below are curl examples for each major endpoint.

## Health Check

Endpoint: GET /api/v1/health

```
curl -s http://localhost:8000/api/v1/health | jq .
```

## List Companies

Endpoint: GET /api/v1/companies

```
curl -s http://localhost:8000/api/v1/companies | jq .
```

## Company Profile

Endpoint: GET /api/v1/companies/{company\_id}

```
curl -s http://localhost:8000/api/v1/companies/TCS | jq .
```

## Financials

Endpoint: GET /api/v1/companies/{company\_id}/financials

```
curl -s http://localhost:8000/api/v1/companies/TCS/financials | jq .
```

## Ratios

Endpoint: GET /api/v1/companies/{company\_id}/ratios

```
curl -s http://localhost:8000/api/v1/companies/TCS/ratios | jq .
```

## Cashflow

Endpoint: GET /api/v1/companies/{company\_id}/cashflow

```
curl -s http://localhost:8000/api/v1/companies/TCS/cashflow | jq .
```

## Peers

Endpoint: GET /api/v1/companies/{company\_id}/peers

```
curl -s http://localhost:8000/api/v1/companies/TCS/peers | jq .
```

## Pros & Cons

Endpoint: GET /api/v1/companies/{company\_id}/pros-cons

```
curl -s http://localhost:8000/api/v1/companies/TCS/pros-cons | jq .
```

## Documents

Endpoint: GET /api/v1/companies/{company\_id}/documents

```
curl -s http://localhost:8000/api/v1/companies/TCS/documents | jq .
```

## Screener

Endpoint: GET

/api/v1/screener?min\_roe=15&sort;=return\_on\_equity\_pct&sort;\_dir=desc&page;=1&page;\_size=20

```
curl -s 'http://localhost:8000/api/v1/screener?min_roe=15&sort;=return_on_equity_pct&sort;_dir=desc' | jq .
```

## Sector Aggregates

Endpoint: GET /api/v1/sectors

```
curl -s http://localhost:8000/api/v1/sectors | jq .
```

# Troubleshooting

This section covers common issues and their solutions.

## Dashboard not loading

Ensure the virtual environment is activated and all dependencies are installed. Run 'pip install -r requirements.txt' to install or update dependencies.

## Database not found

The database file db/nifty100.db must exist. Run the ETL pipeline to create it: 'python src/etl/loader.py'.

## Company list empty

The database may not have data loaded yet. Run the ETL pipeline and verify the companies table has records.

## Port conflicts

If localhost:8501 or localhost:8000 is in use, pass a different port: 'streamlit run src/dashboard/app.py --server.port 8502' or 'uvicorn src.api.main:app --port 8001'.

## PDF tearsheet generation failed

Ensure the company\_id exists in the database. Check the output directory exists and is writable.

## Streamlit charts not rendering

Ensure Plotly and Streamlit are installed and up to date. Clear the Streamlit cache: 'rm -rf .streamlit/cache'.

## API CORS errors

The API allows all origins by default. If you see CORS errors behind a proxy, configure your reverse proxy to pass through headers.

## Screener filters returning empty

Filters use strict min/max bounds. Verify the filter values are within the range of the data. The screener uses the most recent year available.

# Operational Reference

## Testing

Run the test suite with pytest:

```
pytest tests/api -v
pytest tests/analytics -v
pytest tests/ -v
```

## ETL Pipeline

The ETL pipeline loads data from Excel files in data/raw/ into db/nifty100.db:

```
python src/etl/loader.py
```

## Code Quality

Format code with Black and lint with Ruff:

```
black src/ tests/
ruff check src/ tests/
```

## Database Schema

The database is defined in db/schema.sql and contains tables for companies, sectors, financial ratios, market cap, profit and loss, balance sheet, cashflow, peer groups, documents, and pros/cons.

## Useful Commands

| Command         | Description                                                                                | Syntax |
|-----------------|--------------------------------------------------------------------------------------------|--------|
| Start Dashboard | streamlit run src/dashboard/app.py                                                         |        |
| Start API       | uvicorn src.api.main:app --reload                                                          |        |
| Run ETL         | python src/etl/loader.py                                                                   |        |
| Run Tests       | pytest tests/ -v                                                                           |        |
| Format Code     | black src/ tests/                                                                          |        |
| Lint Code       | ruff check src/ tests/                                                                     |        |
| API Docs        | <a href="http://localhost:8000/docs">http://localhost:8000/docs</a>                        |        |
| API Health      | curl <a href="http://localhost:8000/api/v1/health">http://localhost:8000/api/v1/health</a> |        |